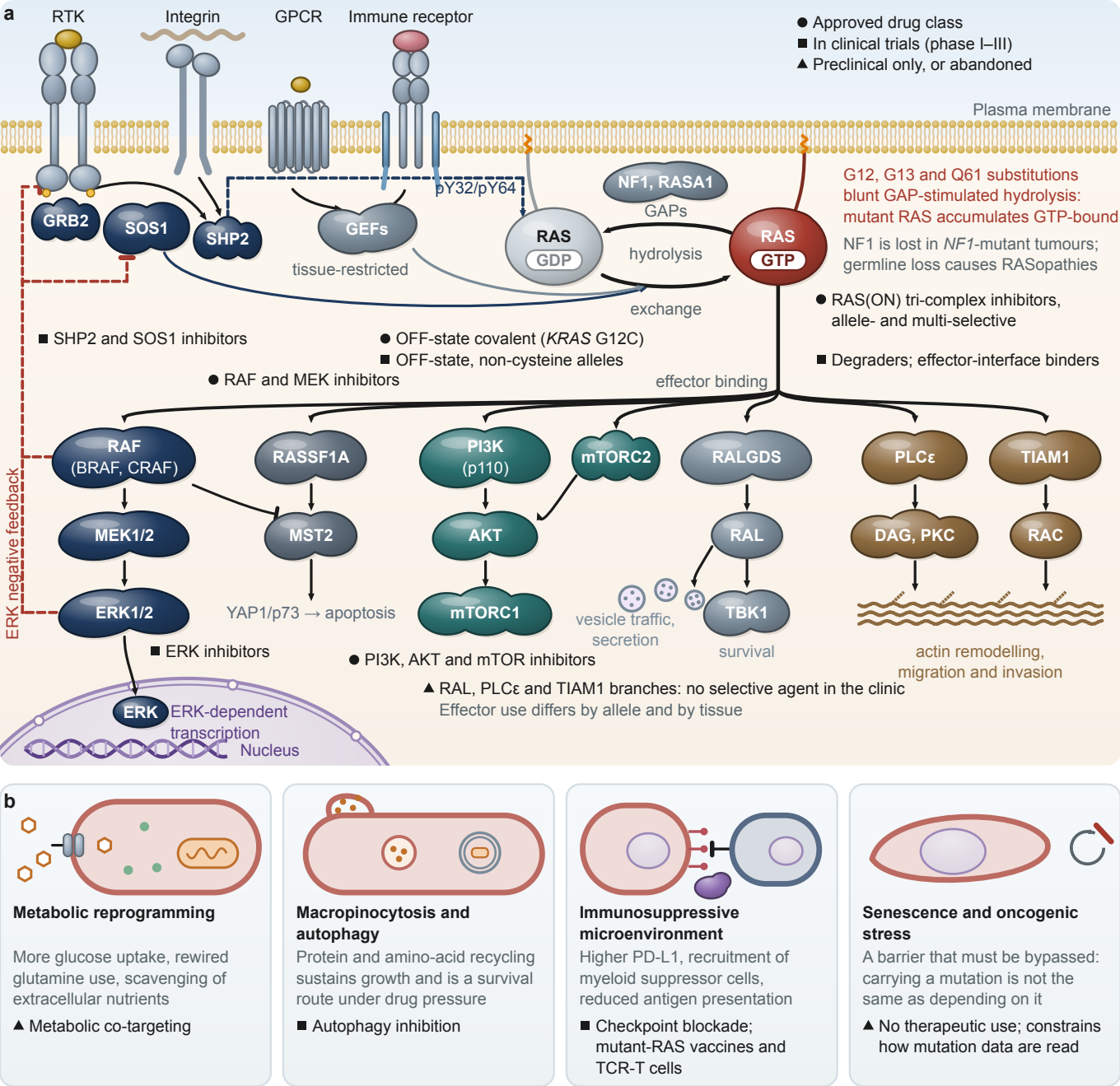

a
RTK
Integrin
GPCR
Immune receptor
● Approved drug class
■ In clinical trials (phase I–III)
▲ Preclinical only, or abandoned
Plasma membrane
GRB2
SOS1
SHP2
GEFs
tissue-restricted
pY32/pY64
RAS
GDP
NF1, RASA1
GAPs
hydrolysis
RAS
GTP
exchange
G12, G13 and Q61 substitutions blunt GAP-stimulated hydrolysis: mutant RAS accumulates GTP-bound
NF1 is lost in NF1-mutant tumours; germline loss causes RASopathies
● RAS(ON) tri-complex inhibitors, allele- and multi-selective
■ Degraders; effector-interface binders
■ SHP2 and SOS1 inhibitors
● OFF-state covalent (KRAS G12C)
■ OFF-state, non-cysteine alleles
● RAF and MEK inhibitors
effector binding
RAF (BRAF, CRAF)
MEK1/2
ERK1/2
ERK negative feedback
RASSF1A
MST2
YAP1/p73 → apoptosis
PI3K (p110)
mTORC2
AKT
mTORC1
RALGDS
RAL
TBK1
vesicle traffic, secretion
survival
PLCε
DAG, PKC
TIAM1
RAC
actin remodelling, migration and invasion
■ ERK inhibitors
● PI3K, AKT and mTOR inhibitors
▲ RAL, PLCε and TIAM1 branches: no selective agent in the clinic
Effector use differs by allele and by tissue
ERK
ERK-dependent transcription
Nucleus
b
Metabolic reprogramming
More glucose uptake, rewired glutamine use, scavenging of extracellular nutrients
▲ Metabolic co-targeting
Macropinocytosis and autophagy
Protein and amino-acid recycling sustains growth and is a survival route under drug pressure
■ Autophagy inhibition
Immunosuppressive microenvironment
Higher PD-L1, recruitment of myeloid suppressor cells, reduced antigen presentation
■ Checkpoint blockade; mutant-RAS vaccines and TCR-T cells
Senescence and oncogenic stress
A barrier that must be bypassed: carrying a mutation is not the same as depending on it
▲ No therapeutic use; constrains how mutation data are read